QA MTH632 QUIZ 1+GRAND QUIZ 100% VERIFIED BY W.S

NEED ONLY PRAYERS

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MTH632:Quiz#1

Quiz Start Time: 05:56 PM, 13 June 2019

Question # 1 of 10 (Start time: 05:56:24 PM, 13 June 2019)

Total Marks: 1

The derivative of f at z_0 i.e. $f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ provided the limit exists. The function f must be defined is a neighbourhood of _____.

Select the correct option

- ○ z
- ● z_0

f must be differentiable not only at $z0$, but also at all points in some ε neighborhood of $z0$.

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MTH632:Quiz#1

Quiz Start Time: 05:56 PM, 13 June 2019

Question # 2 of 10 (Start time: 05:57:15 PM, 13 June 2019)

Total Marks: 1

If a function f(z) = u(x,y) + iv(x,y) is analytic in a domain D, then its component functions u and v are _____ in D.

then its component functions u and v are harmonic function in D.

Select the correct option

- ○ equal
- ○ steady
- ● harmonic
- ○ homogeneous

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MTH632:Quiz#1

Quiz Start Time: 05:56 PM, 13 June 2019

Question # 3 of 10 (Start time: 05:57:27 PM, 13 June 2019)

Total Marks: 1

If Cauchy – Riemann equations are not satisfied at a point z_0 then function is not _____ at z_0.

If Cauchy-Riemann equations equations are not satisfied at z0 then function is not differentiable at z0.

Select the correct option

Reload Math Equations

- integrable
- differentiable

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MTH632:Quiz#1 Quiz Start Time: 10:11 PM, 13 June 20

Question # 9 of 10 (Start time: 10:13:26 PM, 13 June 2019) Total Marks

If f(z) = x + ay + i(bx + cy) is analytic, then a, b, c equals to _____.

Select the correct option

- ○ a= 1 and c= -b
- ○ a = b = c = 1
- ○ b = 1 and a = -c
- ● c=1 and a= -b ✓

$f(z) = x + ay + i(bx + cy)$

$u(x, y) = x + ay \qquad v(x, y) = bx + cy$

$u(x) = 1, u(y) = a \qquad v(x) = b, v(y) = c$

By reimman equation,

$u_x = v_y \qquad u_y = -v_x$

$1 = c \qquad a = -b$

$c = 1$

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MTH632:Quiz#1

Question # 5 of 10 (Start time: 05:58:16 PM, 13 June 2019)

In the complex valued function z = x - iy, the value of U_x = _____.

Select the correct option

- ○ -1
- ○ 0
- ● 1
- ○ 2

$z = x - iy$

$f(z) = \overline{z} = \overline{x} - \overline{i}y = x + iy$

$f(z) = x + iy$

$u(x, y) = x \qquad v(x, y) = y$

$u_x = 1, u_y = 0 \qquad v_x = 0, v_y = 1$

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MTH632:Quiz#1

Question # 7 of 10 (Start time: 05:58:58 PM, 13 June 2019)

In the complex valued function z = x - iy, the value of V_z = _____.

Select the correct option

○	-1
●	0
○	1
○	2

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MTH632:Quiz#1

Quiz Start Time: 05:56 PM, 13 June 2019

Question # 9 of 10 (Start time: 05:59:28 PM, 13 June 2019)

Total Marks: 1

The Cauchy - Riemann equations on a pair of real - valued functions of two real variables $u(x, y)$ and $v(x, y)$ are $U_x = V_y$ and _____.

Select the correct option

Reload Math Equations

- $U_y = -V_y$
- $U_x = -V_x$
- $U_y = -V_x$
- $U_x = -V_y$

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MTH632:Quiz#1

Quiz Start Time: 05:56 PM, 13 June 2019

Question # 10 of 10 (Start time: 05:59:52 PM, 13 June 2019)

Total Marks: 1

A function f(z) is analytic function if _____ part f(z) is analytic.

Select the correct option

- (a) real
- (b) imaginary
- (c) both (a) and (b).
- (d) none of these.

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MTH632 Quiz # 1

Question # 9 of 10 (Start time: 07:48:47 PM, 25 May 2021)

The fifth roots of -1 are exactly

Select the correct option

- cos(± π/10), cos(±3π/10), and -1
- cos(± π/5), cos(±3π/5), and -1 ✓
- cos(π/5), cos(3π/5), cos((-3π)/5) and cos((-π)/5)
- none

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MTH632 Quiz # 1

Question # 8 of 10 (Start time: 08:40:08 PM, 25 May 2021)

If $f(z) = z^3$, $g(z) = \frac{1}{z}$, then $g(f(i)) =$

_____

Select the correct option

- O i^2
- O i^3
- O -1
- ✓ i

$f(z) = z^3, g(z) = \frac{1}{z}$ then $g(f(i)) =$

$f(i) = i^3 = i^2.i = -1.i = -i$

$g(z) = \frac{1}{z}$

$g(f(i)) = \frac{1}{f(i)} = \frac{1}{-i} \times \frac{i}{i} = \frac{i}{-i^2} = \frac{i}{-(-1)} = i$

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MTH632 Quiz # 1

Question # 3 of 10 (Start time: 07:46:27 PM, 25 May 2021)

If $f(x) = x^3$, $g(x) = x + 2$, then $g(f(5)) =$

_____

Select the correct option

○	130
○	7
○	125
●	127

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MTH632 Quiz # 1

Question # 4 of 10 (Start time: 08:19:52 PM, 25 May 2021)

If we write $\Delta z = z - z_0$, then the equation $f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ can be expressed as $f'(z_0) =$ _____.

Select the correct option

- $\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z)}{\Delta z}$
- $\lim_{\Delta z \to 0} \frac{f(\Delta z) - f(z_0)}{\Delta z}$
- $\lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$
- $\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$

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MTH632 Quiz # 1

Question # 1 of 10 (Start time: 07:44:14 PM, 25 May 2021)

If $f(x) = x^3$, $g(x) = x + 2$, then $g(f(4)) =$

_____

Select the correct option

- 4
- 66
- 64
- 16

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MTH632:Quiz # 1

Question # 10 of 10 (Start time: 08:27:07 AM, 25 May 2021)

In general, $1^{a}(1/n)$ is

Select the correct option

O	$\exp\left(i\frac{2k\pi}{n}\right),\ k=0,1,\ldots,n-1$ ✓
O	$\exp\left(i\frac{2\pi}{n}\right)$

S KHAN ACADEMY

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MTH632:Quiz # 1 Quiz Start Time: 08:17 AM, 25 May 2021

Question # 6 of 10 (Start time: 08:23:17 AM, 25 May 2021) Total Marks: 1

The distance between two complex numbers z_1 and z_2 is

Select the correct option

Reload Math Equations

O	$\lvert z_1 - z_2 \rvert$ ✓
O	$\lvert z_1 + z_2 \rvert$

S KHAN ACADEMY

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Quiz Start Time: 08:17 AM, 25 May 202

Question # 8 of 10 (Start time: 08:25:18 AM, 25 May 2021)

Total Marks

A closed disc with center z_0 and radius ε is defined by

Select the correct option

Reload Math Equations

○	$\lvert z_1 - z_0 \rvert \geq \varepsilon$
○	$\lvert z_1 - z_0 \rvert \leq \varepsilon$ ✓

S KHAN ACADEMY

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Question # 2 of 10 (Start time: 08:19:04 AM, 25 May 2021)

If $f(x) = x^2$, $g(x) = \frac{1}{x}$, then $g(f(3)) =$ _____.

Select the correct option

○	1/3
○	3
○	1/9 ✓ S KHAN ACADEMY
○	9

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Question # 4 of 10 (Start time: 08:21:20 AM, 25 May 2021)

The three cube roots of the number -8i are

Select the correct option

○	$2\exp\left[i\left(-\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right],\ k=0,1,2$ ✓
○	$2\exp\left[i\left(-\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right],\ k=0,1$

S KHAN ACADEMY

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Question # 2 of 5 (Start time: 03:38:19 PM, 21 December 2021)

Consider $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$. Then $\lim_{z \to z_0} (f(z)g(z)) =$ _____.

Select the correct option

- ○ $A * B$
- ○ $A.B$
- ○ $A - B$
- ○ $A \pm B$

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Question # 4 of 5 (Start time: 03:55:12 PM, 22 December 2021)

What is the modulus of the complex number 6+0i?

Select the correct option

- -6
- 6
- 0
- None of these

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Quiz Start Time: 08:24 PM, 21 December 2021

Question # 3 of 5 (Start time: 08:25:55 PM, 21 December 2021)

Total Marks: 1

Consider $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$. Then $\lim_{z \to z_0} (f(z) \pm g(z)) =$ _____.

Select the correct option

Reload Math Equations

- $A \pm B$
- $A - B$
- A
- $A + B$

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Question # 3 of 5 (Start time: 11:37:10 AM, 19 December 2021)

Total Marks: 1

$$|z_1 - z_2| =$$

Select the correct option

Reload Math Equations

○	$> \|z_1\| + \|z_2\|$
○	$> z_1 + z_2$
○	$\leqslant z_1 + z_2$
◉	$\leqslant \|z_1\| + \|z_2\|$

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sec(s)

MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1) Quiz Start Time: 02:06 AM, 21 December 2021

Question # 4 of 5 (Start time: 02:09:25 AM, 21 December 2021) Total Marks: 1

The complex number -1-i has exponential form

Select the correct option

Reload Math Equations

- $\sqrt{2}\left[i\left(-\frac{3\pi}{4}\right)\right]$
- $\sqrt{2}\left[i\left(-\frac{3\pi}{2}\right)\right]$

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Question # 5 of 5 (Start time: 11:38:51 AM, 19 December 2021)

Consider $\lim_{z \to z_0} f(z) = A$ *and* $\lim_{z \to z_0} g(z) = B$. *Then* $\lim_{z \to z_0} \frac{f(z)}{g(z)} =$ _____.

Select the correct option

○	$A - B$
◉	$\frac{A}{B}, \ B \neq 0$
○	$A + B$
○	$A \, . \, B$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 07:47:31 PM, 21 December 2021)

if $-8i = 8\exp[i(-\pi/2+2k\pi)]$, then the root of

$(-8i)^{1/3}$

for k=0 is _____

Select the correct option

- $2(\sqrt{3}+i)$
- $\sqrt{3}-i$ ✓
- $\sqrt{3}+i$
- $2(\sqrt{3}-i)$

$$-8i = 8e^{(i(\frac{-\pi}{2}+2k\pi))}$$

$$(-8i)^{\frac{1}{3}} = \left[8e^{(i(\frac{-\pi}{2}+2k\pi))}\right]^{\frac{1}{3}}$$

$$= (2^3)^{\frac{1}{3}} e^{i\left(\frac{-\pi}{2}\times\frac{1}{3}+2k\pi\times\frac{1}{3}\right)}$$

$$= 2e^{i\left(\frac{-\pi}{6}+\frac{2k\pi}{3}\right)}$$

$$= 2e^{i\left(\frac{-\pi+4k\pi}{6}\right)}$$

For $k = 0$

$$(-8i)^{\frac{1}{3}} = 2e^{i\left(\frac{-\pi+0}{6}\right)}$$

$$= 2\left(\cos\left(\frac{-\pi}{6}\right)+i\sin\left(\frac{-\pi}{6}\right)\right)$$

$$= 2\left(\cos\frac{\pi}{6}-i\sin\frac{\pi}{6}\right)$$

$$= 2\left(\frac{\sqrt{3}}{2}-i\frac{1}{2}\right)$$

$$= 2\left(\frac{\sqrt{3}-i}{2}\right)$$

$$= \sqrt{3}-i$$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1) **Quiz Start Time: 02:06 AM, 21 December 2021**

Question # 3 of 5 (Start time: 02:08:35 AM, 21 December 2021) **Total Marks: 1**

Let z = 3, then arg(z) = _____.

Select the correct option

Reload Math Equations

- (●) 270 degree
- ( ) 60 degree
- ( ) 90 degree
- ( ) 0 degree ✓

$$z = 3$$

$$x = 3,\ y = 0$$

$$Arg(z) = \tan^{-1}\left(\frac{y}{x}\right) = 0$$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 07:46:35 PM, 21 December 2021)

Domain of the function

$$f(z) = \frac{1}{z^2 - 1}$$

is _____

Select the correct option

domain of the functions:

$\frac{1}{z^2+1}, z \neq \pm i$

$\frac{1}{z^2-1}, z \neq \pm 1$

$\frac{1}{z^2}, z = 0$

$\frac{1}{z+\bar{z}}, z \neq \pm i$

- $z \neq \pm i$
- $z \neq \pm 1$ ✓
- $z \neq 0$
- $z \neq \pm 2$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 03:56:28 PM, 21 December 2021)

If $-8i = 8\exp[\,i(-\pi/2+2k\pi)\,]$, then the root of $(-8i)^{1/3}$ for $k=2$ is _____

Select the correct option

○	$-\sqrt{3}-i$
○	$-\sqrt{3}+i$
○	$\sqrt{3}-i$
○	$\sqrt{3}+i$

$$-8i = 8e^{(i(\frac{-\pi}{2}+2k\pi))}$$

$$(-8i)^{\frac{1}{3}} = \left[8e^{(i(\frac{-\pi}{2}+2k\pi))}\right]^{\frac{1}{3}}$$

$$= (2^3)^{\frac{1}{3}} e^{i\left(\frac{-\pi}{2}\times\frac{1}{3}+2k\pi\times\frac{1}{3}\right)}$$

$$= 2e^{i\left(\frac{-\pi}{6}+\frac{2k\pi}{3}\right)}$$

$$= 2e^{i\left(\frac{-\pi+4k\pi}{6}\right)}$$

For $k = 2$

$$(-8i)^{\frac{1}{3}} = 2e^{i\left(\frac{-\pi+8\pi}{6}\right)}$$

$$= 2e^{i\left(\frac{7\pi}{6}\right)}$$

$$= 2\left(\cos\frac{7\pi}{6}+i\sin\frac{7\pi}{6}\right)$$

$$= 2\left(\frac{-\sqrt{3}}{2}+i\left(-\frac{1}{2}\right)\right)$$

$$= 2\left(\frac{-\sqrt{3}-i}{2}\right)$$

$$= -\sqrt{3}-i$$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 03:47:57 PM, 21 December 2021)

$$\lim_{(x,y)\to(1,2)} 3xy^2 - y = _____$$

Select the correct option

○	8
○	10
○	9
○	11

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 03:40:01 PM, 21 December 2021)

Sum of complex numbers (3 + 5i) and (4 - 3i) is _____

Select the correct option

- ○ 7 + 2i
- ○ 1+5i
- ○ 7-3i
- ○ 7+5i

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1) Quiz Start Time: 03:37 PM, 21 December 2021

Question # 4 of 5 (Start time: 03:41:02 PM, 21 December 2021) Total Marks: 1

The third roots of 2i is

Select the correct option

Reload Math Equations

- $\sqrt{3}\left[i\left(\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right], k \in Z$
- $\sqrt[3]{2}\left[i\left(\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right], k \in Z$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 03:37:52 PM, 21 December 2021)

The complex number

$$(4i)^4$$

have _____ number of roots

Select the correct option

○	2
○	1
○	4
○	3

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Question # 2 of 5 (Start time: 03:28:09 PM, 21 December 2021)

Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$, then choose the correct option.

Select the correct option

○	$\lim_{z \to z_0} [f(z) + g(z)] = A + B$
○	$\lim_{z \to z_0} [f(z) + g(z)] = \frac{A}{B}$
○	$\lim_{z \to z_0} [f(z) + g(z)] = A \cdot B$
○	$\lim_{z \to z_0} [f(z) + g(z)] = A - B$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 12:04:20 AM, 01 February 2022)

The x - axis is called _____.

Select the correct option

○	an imaginary axis
○	a real axis

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 12:05:38 AM, 01 February 2022)

Conjugate of complex number 8 + 3 i is _____.

Select the correct option

○	-8 - 3i
○	-8 + 3i
○	8 + 3i
○	8 - 3i ✓ ✓

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MTH632 – Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 12:23:50 AM, 01 February 2022)

Difference of complex numbers (4 + 2i) and (1 + 8i) is _____.

Select the correct option

○	(3 - 10i)
○	(5 - 6i)
○	(5 - 10i)
○	(3 - 6i) ✓ ✓

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Question # 4 of 5 (Start time: 12:25:27 AM, 01 February 2022)

If initial point and terminal points are equal then the curve C is said to be_____.

If initial point and terminal points are equal then the curve c is said to be closed curve.

Select the correct option

- ○ closed curve ✓ ✓
- ○ parallel curve
- ○ open curve
- ○ simple curve

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 12:27:02 AM, 01 February 2022)

Product of complex numbers (3 + 2i) and (1+7i) is _____.

Select the correct option

○	3+9i
○	4+9i
○	-11+23i ✓ ✓
○	3+14i

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Question # 3 of 5 (Start time: 09:32:10 AM, 01 February 2022)

The solution of the equation $x^2 + x + 1 = 0$ lies in the set R of real numbers.

Select the correct option

- ○ True
- ◉ False

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 08:38:48 PM, 01 February 2022)

A real number x can be displayed as a point _____ on real axis.

Select the correct option

○	(1, x)
○	(0, x)
○	(x, 1)
◉	(x, 0)  

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 08:39:09 PM, 01 February 2022)

$z_1 = (1, 3)$ and $z_2 = (3, 1)$ are _____.

Select the correct option

◉	unequal
○	equal

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 08:41:44 PM, 01 February 2022)

The y - axis is called _____.

Select the correct option

○	a real axis
◉	an imaginary axis

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 08:41:56 PM, 01 February 2022)

The commutative law shows that _____.

Select the correct option

○	$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$
○	$z_1 (z_2 + z_3) = (z_1 z_2 + z_1 z_3)$
◉	$z_1 + z_2 = z_2 + z_1$
○	$z + w = z$

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MTH632 – Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 08:42:38 PM, 01 February 2022)

Domain of the function

$$f(z) = \frac{1}{z^2}$$

Is _____.

Select the correct option

○	$z \neq \pm 2$
◉	$z \neq \pm i$
○	$z \neq \pm 1$
○	$z \neq 0$ ✓ ✓

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 08:51:02 PM, 01 February 2022)

Let z = 7 i, then Arg(z) = _____.

Select the correct option

$z = 7i$

$z = 0 + 7i$

$x = 0,\ y = 7$

$Arg(z) = 90^0 = \frac{\pi}{2}$

○	$\frac{\pi}{3}$
○	$\frac{\pi}{4}$
◉	$\frac{\pi}{2}$
○	$\frac{\pi}{6}$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 08:51:23 PM, 01 February 2022)

Let $z = 1 + i$, then $Arg(z) =$ _____.

Select the correct option

○	$2\pi/4$
◉	$\pi/4$
○	$3\pi/4$
○	$4\pi/4$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 08:51:44 PM, 01 February 2022)

The additive identity of a complex number z = (x, y) is _____.

Select the correct option

○	(1, 0)
◉	(0, 0)

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 08:52:30 PM, 01 February 2022)

The conjugate of 6 + 3i is

Select the correct option

○	-6-3i
◉	6-3i
○	-6+3i
○	6+3i

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 09:05:03 PM, 01 February 2022)

Conjugate of complex number $-7 - 4i$ is _____.

Select the correct option

◉	$-7 + 4i$
○	$7 - 4i$
○	$7 + 4i$
○	$-7 - 4i$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 09:05:20 PM, 01 February 2022)

What is the modulus of the complex number -6 +8i?

Select the correct option

○	-10
◉	10
○	14
○	6

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 09:05:45 PM, 01 February 2022)

What is the modulus of the complex number 8i?

Select the correct option

○	-8
○	i
◉	8
○	-i

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 09:06:01 PM, 01 February 2022)

The solution of the equation $x^2 + 1 = 0$ lies in the set of complex numbers.

Select the correct option

◉	True
○	False

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Question # 1 of 5 (Start time: 09:21:18 PM, 01 February 2022)

The polar form of a complex number z is _____.

Select the correct option

○	$r(sec\theta + i\cos ec\theta)$
◉	$r(cos\theta + i\sin\theta)$
○	$r(tan\theta + i\cot\theta)$
○	$r(sin\theta + i\cos\theta)$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 09:22:28 PM, 01 February 2022)

The equation $x^2 - 2x + 3 = 0$ does not have the real roots.

Select the correct option

◉	False
○	True

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 09:23:50 PM, 01 February 2022)

Let z = 1 + i, then r = _____.

Select the correct option

○	2
○	$\sqrt{3}$
◉	$\sqrt{2}$ ✓ ✓
○	1

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 09:24:07 PM, 01 February 2022)

The additive inverse of (5, 7) is _____.

Select the correct option

○	(5, -7)
○	(-5, 7)
◉	(-5, -7)
○	(5, 7)

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 09:49:40 PM, 01 February 2022)

$$|z_1 + z_2| =$$

Select the correct option

○	$> z_1 + z_2$
○	$> |z_1| + |z_2|$
◉ ✓	$\leqslant |z_1| + |z_2|$ ✓
○	$\leqslant z_1 + z_2$

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Question # 2 of 5 (Start time: 09:50:29 PM, 01 February 2022)

The limit $\lim_{z \to 0} \frac{z}{\bar{z}}$ _____.

Select the correct option

○	exists
◉	does not exists

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Question # 3 of 5 (Start time: 09:50:49 PM, 01 February 2022)

In z = a + bi, if i is replaced by -i, then another complex number obtained is said to be

Select the correct option

○	multiplicative inverse of z
○	prime factor of z
◉	Complex conjugate of z
○	additive inverse of z

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 09:51:05 PM, 01 February 2022)

$If\ z = a + ib\ is\ a\ complex\ number,\ then$

Select the correct option

○	None of these
○	$\mathrm{Re}(z) = \frac{z+\bar{z}}{2}$ ✓
○	$\mathrm{Re}(z) = \frac{z+\bar{z}}{2i}$
○	$\mathrm{Re}(z) = z + \bar{z}$

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MTH632 – Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 09:52:14 PM, 01 February 2022)

The Jordan Curve Theorem guarantees that a simple closed curve must enclose a region.

Select the correct option

○	Euler curve
○	Simple curve
○	Jordan curve
○	Rotation curve

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Question # 1 of 5 (Start time: 10:23:01 PM, 01 February 2022)

The function $u(x,y)=\frac{xy}{x^2+y^2}$ does not have a limit when _____.

Select the correct option

○	$(x,y)\to(1,0)$
○	$(x,y)\to(1,1)$
○	$(x,y)\to(0,1)$
◉	$(x,y)\to(0,0)$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 10:23:21 PM, 01 February 2022)

The Cartesian form of the complex number $6\text{cis}(\pi)$ is

Select the correct option

○	-6π
○	6π
○	6
◉	-6

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 10:23:32 PM, 01 February 2022)

A curve is called _____ if it does not cross itself.

A curve is called simple if it does not cross itself.

Select the correct option

○	open curve
○	parallel curve
○	closed curve
◉	simple curve

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MTH632 – Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 10:24:14 PM, 01 February 2022)

A complex power function is a function of the form _____, where

α is a complex constant.

Complex power function is a function of the form $f(z)=z\alpha$ where α is a complex constant.

Select the correct optio

◉	✔	$f(z) = z^{\alpha}$ ✓
○		$f(\alpha) = z^{\alpha}$
○		$f(\alpha) = \alpha^{z}$
○		$f(z) = \alpha^{z}$

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MTH632 – Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 10:24:27 PM, 01 February 2022)

$$(1+i)^2 = _____$$

Select the correct option

○	1+i
○	1−2i
○	1+2i
◉	0+2i

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 10:26:06 PM, 01 February 2022)

If (2, 3) + (a, b) = (0, 0), then (a, b) = _____.

Select the correct option

○	(-2, 3)
◉	(-2, -3)
○	(2, -3)
○	(2, 3)

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 10:26:23 PM, 01 February 2022)

The representation $a + i\,b$ of a complex number z is due to Gauss.

Select the correct option

◉	True
○	False

Most of the time we use the following representation of complex numbers $z=x+iy$ The representation is due to Gauss.

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 10:27:04 PM, 01 February 2022)

$Re(z)$ of $z = (5, 7) + (3, -2)$ is _____.

Select the correct option

◉	8
○	3
○	7
○	5

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 10:34:37 PM, 01 February 2022)

The root of the equation $6x + 1 = 3$ is _____.

Select the correct option

○	an irrational number.
◉	a rational number

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 10:34:51 PM, 01 February 2022)

Mapping is _____ way of visualizing a given function.

Select the correct option

○	a numerical
○	none of the above
○	an analytical
◉	a geometrical

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 10:35:02 PM, 01 February 2022)

If z = -1+i, then z expressed in polar form is

Select the correct option

◉	$2cis\left(-\frac{\pi}{4}\right)$ ✓
○	$cis\left(-\frac{\pi}{4}\right)$
○	$2cis\left(\frac{3\pi}{4}\right)$
○	$cis\left(\frac{3\pi}{4}\right)$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 10:35:38 PM, 01 February 2022)

$$Let\ z\overline{z} = 64, then\ |z| =$$

Select the correct option

○	-8
◉	8
○	64
○	-64

$z\bar{z} = |z|^2$

$|z| = 8$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 10:42:18 PM, 01 February 2022)

$Im(z)$ of $z = (5, 7) + (3, -2)$ is _____.

Select the correct option

◉	5
○	7
○	8
○	3

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 10:42:43 PM, 01 February 2022)

Sum of complex numbers (3 + 2i) and (1 + 7i) is _____.

Select the correct option

○	3+14i
◉	4 + 9i
○	3+7i
○	3+9i

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Quiz Start Time: 10:42 PM, 01 February 2022

Question # 3 of 5 (Start time: 10:42:59 PM, 01 February 2022)

Total Marks: 1

If f approaches two complex numbers $L_1 \neq L_2$ for two different curves or paths through z_0 then the $\lim_{z \to z_0} f(z)$ _____.

different curves or paths through z, then lim $z \to z$ $f\,z$ does not exist.

Select the correct option

Reload Math Equations

- (●) does not exist
- ( ) exists

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 5 of 5 (Start time: 10:43:26 PM, 01 February 2022)

If $f(z)=z^2$, then $f(x+iy)=$ _____.

$$f(z) = z^2$$
$$f(x+iy) = (x+iy)^2$$
$$= x^2 + i^2y^2 + 2xiy$$
$$= x^2 + (-1)y^2 + 2xyi$$
$$= x^2 - y^2 + 2xyi$$

Select the correct option

◉	$x^2 - y^2 + 2xyi$ ✓ ✓
○	$x^2 - y^2 i - 2xyi$
○	$x^2 + y^2 - 2xyi$
○	$x^2 + y^2 i - 2xyi$

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Question # 2 of 5 (Start time: 11:14:26 PM, 01 February 2022)

Polar form of the complex number 2-2i is

Select the correct option

◉	$2\sqrt{2}cis\left(-\frac{\pi}{4}\right)$ ✓
○	none of these
○	$2cis\left(-\frac{\pi}{4}\right)$
○	$2cis\left(\frac{\pi}{4}\right)$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 3 of 5 (Start time: 11:15:10 PM, 01 February 2022)

The associative law shows that _____.

Select the correct option

- ○ $z + w = z$
- ◉ $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$ ✓ ✓
- ○ $z_1 (z_2 + z_3) = (z_1 z_2 + z_1 z_3)$
- ○ $z_1 + z_2 = z_2 + z_1$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Quiz Start Time: 11:14 PM, 01 Februar

Question # 5 of 5 (Start time: 11:16:45 PM, 01 February 2022)

Total M

A set S is said to be _____ set if for every pair of points z1, z2∈S there is a curve C joining them which is entirely contained in S.

A set $S$is said to be connected set if for every pair of points $z_1z_2 \in S$there is a curve Cjoining them which is entirely contained in S.

Select the correct option

Reload Math Equat

- unbounded
- unconnected
- connected
- bounded

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 12:15:04 PM, 01 February 2022)

A complex - valued function of the form $L(z) = Az + B$, *where* $A, B \in \mathbb{Z}$, represents a _____ transformation.

Select the correct option

Page 398

- quadratic
- cubic
- none of the above
- linear

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (Start time: 12:15:17 PM, 01 February 2022)

By definition, $\lim_{z \to z_0} f(z) = L$ means that for every $\varepsilon > 0$ there exists a $\delta > 0$ such that _____.

Page 539

Select the correct option

- $|f(z) - z_0| < \varepsilon$ whenever $0 < |z - L| < \delta$
- $|f(z) - z_0| < \delta$ whenever $0 < |z - L| < \varepsilon$
- $|f(z) - L| < \delta$ whenever $0 < |z - z_0| < \varepsilon$
- $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 12:17:31 PM, 01 February 2022)

If two complex - valued functions f and g satisfy conditions $g(f(z)) = z$ and $f(g(w))$ then these functions are_____of each other.

Select the correct option

- ○ none of the above
- ○ conjugates
- ○ reciprocals
- ◉ inverses

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 04:45:50 PM, 01 February 2022)

In an Argand diagram, the complex number z = 2i will be a point _____.

Select the correct option

○	2 units along the positive real axis
○	2 units along the positive imaginary axis ✓ ✓

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 11:11:34 PM, 01 February 2022)

The principal argument of -5-5i is

Select the correct option

○	$\frac{-3\pi}{4}$	
○	$\frac{5\pi}{4}$	✓
○	$-\frac{5\pi}{4}$	
○	$\frac{3\pi}{4}$	

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 10:00:33 PM, 01 February 2022)

A complex-valued linear transformation is always _____ on the entire complex plane.

Select the correct option

- none of the above
- only onto
- only one-to-one
- one-to-one and onto ✓

Inverse Translation

Translation is a one-to-one and onto function on the entire complex plane.

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 4 of 5 (Start time: 12:13:10 PM, 02 February 2022)

Which of the following does NOT describe the modulus of the complex number a + bi?

Select the correct option

◉	The difference between the numbers a and b.
○	$\sqrt{a^2 + b^2}$
○	The magnitude of the vector representing the complex number a + bi.
○	The length of the directed line segment from the origin of the complex plane to the point (a, b).

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 07:59:20 PM, 01 February 2022)

According to _____, $(2 + i3) + (5 + i4) = (5 + i4) + (2 + i3)$

Select the correct option

○	Associative law
○	Commutative law

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rH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

uestion # 2 of 5 (Start time: 08:00:01 PM, 01 February 2022)

Which of the following statements is TRUE?

i think its D but not confirmed

elect the correct option

○	The set Q of rational numbers is an extension of set Q' of irrational numbers.
○	The set R of real numbers is an extension of set C of complex numbers.
○	The set Q' of irrational numbers is an extension of set Z of integers.
○	The set C of complex numbers is an extension of set R of real numbers.

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 2 of 5 (start time: 12:12:35 PM, 02 February 2022)

$\sqrt{2}$ is a rational number.

Select the correct option

- (●) False
- ( ) True

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MTH632 - Complex Analysis and Differential Geometry (Quiz No. 1)

Question # 1 of 5 (Start time: 12:11:12 PM, 02 February 2022)

Domain of the function

$$f(z) = \frac{1}{z^2 + 1}$$

is _____.

Select the correct option

○		$z \neq \pm 2$	
○		$z \neq \pm 1$	
◉	✓	$z \neq \pm i$	✓
○		$z \neq 0$	

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If Cauchy – Riemann equations are satisfied at a point z_0 then function_____ differentiable at z_0

the correct option

(a) may be

(b) may not be

(c) Both (a) and (b).

(d) None of these.

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Question # 10 of 30 (Start time: 10:23:35 AM, 01 July 2020)

Total Marks: 1

The sequence $z_n = \frac{i}{n^3}\ (n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____.

Select the correct option

Reload Math Equations

- O 0 ✓
- O 1/2
- O 1
- O infinity

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MTH632:Grand Quiz

Question # 15 of 30 (Start time: 10:56:40 AM, 01 July 2020)

If $e^{ax}\cos y$ is harmonic then 'a' is _____.

Select the correct option

○	1
○	-1 correct ✓
○	0
○	2

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MTH632:Grand Quiz

Question # 6 of 30 (Start time: 10:05:54 AM, 01 July 2020)

A function V is said to be harmonic if and only if _____.

Select the correct option

○	$V_x + V_y = 0$
○	$V_{xy} + V_{yx} = 0$
○	$V_{xx} + V_{yy} = 0$ ✓
○	$V_x^2 + V_y^2 = 0$

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Question # 30 of 30 (Start time: 10:30:01 AM, 01 July 2020)

Total Marks: 1

Any polynomial $p(z) = a_0 + a_1 z + a_2 z^2 + \ldots + a_n z^n \ (a_n \neq 0)$, of degree $n (n \geq 1)$ has least one zero, i.e. there exists at least _____ z_0, such that $P(z_0) = 0$.

Select the correct option

Reload Math Equations

O	n points
O	10 points
O	one point
O	no points

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MTH632:Grand Quiz

Question # 21 of 30 (Start time: 10:21:32 AM, 01 July 2020)

Domain of the function

$$f(z) = \frac{1}{z^2}$$

Select the correct option

O	$z \neq \pm 1$
O	$z \neq \pm 2$
O	$z \neq \pm i$
O	$z \neq 0$ ✓

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Question # 27 of 30 (Start time: 10:37:49 AM, 01 July 2020)

A complex-valued function of the form

$$L(z) = Az + B,$$

where

Select the correct option

O	linear
O	quadratic
O	cubic
O	none of the above

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Question # 22 of 30 (Start time: 10:22:27 AM, 01 July 2020)

Total

The derivative of f at z_0 i.e. $f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ provided the limit exists. The function f must be defined is a neighbourhood of ___

Select the correct option

Reload Math E

O	z
O	z_0 ✓

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MTH632:Grand Quiz

Quiz Start Time: 10:01 AM, 01 July 2020

Question # 11 of 30 (Start time: 10:11:07 AM, 01 July 2020)

Total Marks: 1

A function f(z) is analytic function if _____ part f(z) is analytic.

Select the correct option

O	(a) real
O	(b) imaginary
O	(c) both (a) and (b).
O	(d) none of these.

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MTH632:Grand Quiz

Question # 16 of 30 (Start time: 10:16:13 AM, 01 July 2020)

Domain of the function

$$f(z) = \frac{1}{z^2 + 1}$$

Select the correct option

O	$z \neq \pm 1$
O	$z \neq \pm 2$
O	$z \neq \pm i$ ✓
O	$z \neq 0$

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Question # 12 of 30 (Start time: 09:44:20 AM, 01 July 2020)

Let z = 3, then arg(z) = _____.

Select the correct option

○	0 degree
○	60 degree
○	90 degree
○	270 degree

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Question # 27 of 30 (Start time: 10:27:17 AM, 01 July 2020)

Consider $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$. Then $\lim_{z \to z_0} (f(z)g(z)) =$ _____.

Select the correct option

O	$A + B$
O	$A - B$
O	$A.B$ ✓
O	$A \pm B$

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MTH632:Grand Quiz

Quiz Start Time: 10:01 AM, 01 July 2020

Question # 10 of 30 (Start time: 10:10:22 AM, 01 July 2020)

Total Marks: 1

If U and V are harmonic functions then $f(Z) = U + iV$ is _____.

Select the correct option

O	analytic function
O	need not be analytic function.
O	analytic function only at Z=0.
O	None of these.

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Question # 26 of 30 (Start time: 10:11:55 AM, 01 July 2020)

Conjugate of complex number 8+3i is _____.

Select the correct option

- 8+3i
- 8-3i
- -8+3i
- -8-3i

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Question # 24 of 30 (Start time: 10:24:12 AM, 01 July 2020)

A connected open set is called a _____.

Select the correct option

- O domain ✓
- O range
- O closed set
- O unbounded set

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MTH632:Grand Quiz

Quiz Start Time: 10:01 AM, 01 July 2020

Question # 2 of 30 (Start time: 10:02:22 AM, 01 July 2020)

Total Marks: 1

The Jordan Curve Theorem guarantees that a simple closed curve must enclose a region.

Select the correct option

- O Euler curve
- O Jordan curve
- O Simple curve ✓
- O Rotation curve

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Question # 17 of 30 (Start time: 10:05:16 AM, 01 July 2020)

If $f(z) = z^3$, $g(z) = z + 2$, then $g(f(4)) =$

_____.

Select the correct option

O	4
O	16
O	64
O	66

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MTH632:Grand Quiz

Quiz Start Time: 09:52 AM, 01 July 2020

Question # 15 of 30 (Start time: 10:03:38 AM, 01 July 2020)

Total Marks: 1

If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____.

Select the correct option

- a= 1 and c= -b
- a = b = c = 1
- b = 1 and a = -c
- c=1 and a= -b

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MTH632:Grand Quiz

Question # 12 of 30 (Start time: 09:49:58 AM, 01 July 2020)

Domain of the function

$$f(z) = \frac{1}{z^2 - 1}$$

Select the correct option

O	$z \neq \pm 1$ ✓
O	$z \neq \pm 2$
O	$z \neq \pm i$
O	$z \neq 0$

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4 of 30 (Start time: 09:45:30 AM, 01 July 2020)

Consider $\lim_{z \to z_0} f(z) = A$ *and* $\lim_{z \to z_0} g(z) = B$. *Then* $\lim_{z \to z_0} (f(z) \pm g(z)) =$ _____.

orrect option

A

$A + B$

$A - B$

$A \pm B$ ✓

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Question # 8 of 30 (Start time: 10:08:26 AM, 01 July 2020)

The sequence $z_n = -3 + i\frac{1}{n^3}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____.

Select the correct option

O	i
O	-1
O	1
O	-3 ✓

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MTH632:Grand Quiz

Question # 25 of 30 (Start time: 11:03:27 AM, 01 July 2020)

In the complex valued function z = x - iy, the value of U_x = _____.

Select the correct option

O	-1
O	0
O	1
O	2

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MTH632:Grand Quiz

Quiz Start Time: 10:16 AM, 01 July 2020

Question # 9 of 30 (Start time: 10:22:36 AM, 01 July 2020)

Total Marks: 1

Mapping is _____ way of visualizing a given function.

Select the correct option

- an analytical
- a geometrical ✓
- a numerical
- none of the above

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MTH632:Grand Quiz

Question # 1 of 30 (Start time: 10:01:16 AM, 01 July 2020)

Domain of the function

$$f(z) = \frac{1}{z + \bar{z}}$$

Select the correct option

O	$z \neq \pm 1$	
O	$z \neq \pm 2$	
O	$z \neq \pm i$	✓
O	$z \neq 0$	

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MTH632:Grand Quiz

Quiz Start Time: 09:52 AM, 01 July 202

Question # 12 of 30 (Start time: 10:00:56 AM, 01 July 2020)

Total Marks:

Sum of complex numbers (3 + 5i) and (4 − 3i) is _____.

Select the correct option

- 1+5i
- 7-3i
- 7+5i
- 7 + 2i ✓

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MTH632:Grand Quiz

Question # 17 of 30 (Start time: 09:47:35 AM, 01 July 2020)

If -8i = 8 exp[i (-π/2+2kπ)], then the root of

$$(-8i)^{1/3}$$

for k=2 is _____.

Select the correct option

○	$\sqrt{3}-i$
○	$\sqrt{3}+i$
○	$-\sqrt{3}-i$
○	$-\sqrt{3}+i$

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Question # 21 of 30 (Start time: 10:32:59 AM, 01 July 2020)

$(1+i)^2 =$ _____.

Select the correct option

O	1+2i
O	0+2i ✓
O	1+i
O	1-2i

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MTH632:Grand Quiz

Question # 14 of 30 (Start time: 10:14:08 AM, 01 July 2020)

If $f(z) = \dfrac{z^4+1}{(z^2+1)(z-3)}$, then f(z) is not analytic at z =

Select the correct option

O	1
O	-1
O	i
O	2

$$f(z) = \frac{z^4+1}{(z^2+1)(z-3)}$$

$$(z^2+1)(z-3) = 0$$

$z^2+1=0$ $\qquad$ $z-3=0$

$z=\pm\sqrt{-1}$ $\qquad$ $z=3$

$z=\pm i$

$f(z)$ *is not analytical at* $z=\pm i, 3$

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MTH632:Grand Quiz

Quiz Start Time: 10:16 AM, 01 July 2020

Question # 4 of 30 (Start time: 10:18:24 AM, 01 July 2020)

Total Marks: 1

A set S is said to be _____ set if for every pair of points z1, z2∈S there is a curve C joining them which is entirely contained in S.

Select the correct option

Reload Math Equations

O	bounded
O	unbounded
O	unconnected
O	connected ✓

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MTH632:Grand Quiz

Question # 7 of 30 (Start time: 10:07:03 AM, 01 July 2020)

A complex power function is a function of the form _____, where

α

is a complex constant.

Select the correct option

O	$f(z) = z^{\alpha}$ ✓
O	$f(z) = \alpha^{z}$
O	$f(\alpha) = z^{\alpha}$
O	$f(\alpha) = \alpha^{z}$

Scanned with CamScanner

MTH632:Grand Quiz

Quiz Start Time: 10:01 AM, 01 July 2020

Question # 26 of 30 (Start time: 10:26:34 AM, 01 July 2020)

Total Marks: 1

If initial point and terminal points are equal then the curve C is said to be_____.

Select the correct option

- o closed curve ✓
- o open curve
- o simple curve
- o parallel curve

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Question # 6 of 30 (Start time: 09:44:05 AM, 01 July 2020)

A complex-valued linear transformation could be a composition of _____.

Select the correct option

- translation
- rotation
- magnification
- all of the above ✓

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MTH632:Grand Quiz

Question # 11 of 30 (Start time: 09:43:41 AM, 01 July 2020)

If $f(z) = z^3$, $g(z) = z + 2$, then $g(f(5)) =$

_____

Select the correct option

○	7
○	125
○	127 ✓
○	130

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Question # 6 of 30 (Start time: 10:19:57 AM, 01 July 2020)

Total Marks: 1

Sum of complex numbers (3 + 2i) and (1 + 7i) is _____.

Select the correct option

O	4 + 9i
O	3+7i
O	3+14i
O	3+9i

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MTH632:Grand Quiz

Question # 9 of 30 (Start time: 10:09:18 AM, 01 July 2020)

If $f(z) = z^3$, $g(z) = \frac{1}{z}$, then $g(f(-1)) =$ ____.

Select the correct option

O	1
O	-1 ✓
O	2
O	-2

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MTH632:Grand Quiz

Quiz Start Time: 10:01 AM, 01 July 2020

Question # 13 of 30 (Start time: 10:12:45 AM, 01 July 2020)

Total Marks: 1

Product of complex numbers (3 + 2i) and (1+7i) is _____.

Select the correct option

O	3+9i
O	4+9i
O	-11+23i ✓
O	3+14i

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MTH632:Grand Quiz

Question # 5 of 30 (Start time: 10:04:59 AM, 01 July 2020)

Let z = 1 + i, then arg(z) = _____.

Select the correct option

○	2π/4
○	3π/4
○	4π/4
○	π/4

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MTH632:Grand Quiz

Question # 1 of 30 (Start time: 08:10:46 AM, 01 July 2020)

Let $z = 1 + i$, then $r =$ _____

Select the correct option

O	$\sqrt{2}$
O	$\sqrt{3}$
O	2
O	1

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MTH632 Grand Quiz

Quiz Start Time: 05:16 A

Question # 20 of 30 (Start time: 05:23:28 AM, 01 July 2020)

Conjugate of complex number -7-4i is _____.

Select the correct option

- -7-4i
- 7-4i
- -7+4i
- 7+4i

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MTH632:Grand Quiz

Question # 9 of 30 (Start time: 08:22:30 AM, 01 July 2020)

If $f(z)=z^2$, then $f(x+iy)=$ _____.

Select the correct option

○	$x^2 - y^2 + 2xyi$
○	$x^2 + y^2 - 2xyi$
○	$x^2 + y^2 i - 2xyi$
○	$x^2 - y^2 i - 2xyi$

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MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 23 of 30 (Start time: 10:08:01 AM, 02 July 2020)

Total Marks: 1

Given a complex - valued function $s(t)$, a rotation by an angle $\frac{\pi}{3}$ is given by _____.

Select the correct option

Reload Math Equations

- $e^{\frac{3\pi}{T}}s(t)$
- $e^{\frac{2\pi}{T}}s(t)$
- correct $e^{\frac{\pi}{T}}s(t)$
- none of the above

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MTH632:Grand Quiz

Quiz Start Time: 09:57 AM, 02 July 2020

Question # 2 of 30 (Start time: 09:57:21 AM, 02 July 2020)

Total Marks: 1

By definition, $\lim_{z \to z_0} f(z) = L$ means that for every $\varepsilon > 0$ there exists a $\delta > 0$ such that _____.

Select the correct option

Reload Math Equations

- ○ $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$ ✓ correc
- ○ $|f(z) - z_0| < \varepsilon$ whenever $0 < |z - L| < \delta$
- ○ $|f(z) - L| < \delta$ whenever $0 < |z - z_0| < \varepsilon$
- ○ $|f(z) - z_0| < \delta$ whenever $0 < |z - L| < \varepsilon$

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MTH632:Grand Quiz

Quiz Start Time: 07:02 AM, 02 July 2020

Question # 17 of 30 (Start time: 07:12:19 AM, 02 July 2020)

Total Marks: 1

Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$, then choose the correct option.

Select the correct option

Reload Math Equations

- $\lim_{z \to z_0}[f(z) + g(z)] = A + B$ ✓ correct
- $\lim_{z \to z_0}[f(z) + g(z)] = A - B$
- $\lim_{z \to z_0}[f(z) + g(z)] = A \cdot B$
- $\lim_{z \to z_0}[f(z) + g(z)] = \frac{A}{B}$

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MTH632:Grand Quiz

Question # 6 of 30 (Start time: 08:18:36 AM, 01 July 2020)

If -8i = 8 exp[i (-π/2+2kπ)], then the root of

root is not given in the question

for k=0 is _____.

Select the correct option

○	$\sqrt{3} - i$ ✓ correct
○	$\sqrt{3} + i$
○	$2(\sqrt{3} - i)$
○	$2(\sqrt{3} + i)$

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MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 11 of 30 (Start time: 08:22:07 AM, 01 July 2020)

Total Marks: 1

If a function f(z) = u(x,y) + iv(x,y) is analytic in a domain D, then its component functions u and v are _____ in D.

Select the correct option

- o equal
- o steady
- o harmonic ✓ correct
- o homogeneous

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MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 9 of 30 (Start time: 08:20:00 AM, 01 July 2020)

Total Marks: 1

If Cauchy – Riemann equations are not satisfied at a point z_0 then function is not _____ at z_0.

Select the correct option

Reload Math Equations

O	integrable
O	differentiable ✓ correct

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MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 5 of 30 (Start time: 08:15:39 AM, 01 July 2020)

Total Marks: 1

Consider $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$. Then $\lim_{z \to z_0} \frac{f(z)}{g(z)} =$ _____.

Select the correct option

Reload Math Equations

○	$\frac{A}{B},\ B \neq 0$ ✓ correct
○	$A + B$
○	$A - B$
○	$A.B$

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MTH632:Grand Quiz

Quiz Start Time: 08:10 AM, 01 July 2020

Question # 3 of 30 (Start time: 08:13:17 AM, 01 July 2020)

Total Marks: 1

The Cauchy - Riemann equations on a pair of real - valued functions of two real variables $u(x, y)$ and $v(x, y)$ are $U_x = V_y$ and _____.

Select the correct option

Reload Math Equations

O	$U_y = -V_y$
O	$U_x = -V_x$
O	$U_y = -V_x$ ✓ correct
O	$U_x = -V_y$

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Question # 2 of 30 (Start time: 08:12:08 AM, 01 July 2020)

If $f(z) = z^3$, $g(z) = \frac{1}{z}$, then $g(f(i)) =$

Select the correct option

O	i
O	i^2
O	i^3
O	-1



correct

$f(z) = z^3, g(z) = \frac{1}{z}$ then $g(f(i)) =$

$f(i) = i^3 = i^2.i = -1.i = -i$

$g(z) = \frac{1}{z}$

$g(f(i)) = \frac{1}{f(i)} = \frac{1}{-i} \times \frac{i}{i} = \frac{i}{-i^2} = \frac{i}{-(-1)} = i$

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MTH632:Grand Quiz

Question # 13 of 30 (Start time: 07:11:22 PM, 02 January 2021)

In the complex valued function z = x - iy, the value of $V_x =$ _____.

Select the correct option

○	-1
○	0 ✓
○	1
○	2

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MTH632:Grand Quiz

Question # 30 of 30 (Start time: 07:21:52 PM, 02 January 2021)

Let z = 7 i, then arg(z) = _____.

Select the correct option

○	π/4
○	π/2 ✓
○	π/3
○	π/6

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Question # 27 of 30 (Start time: 07:20:14 PM, 02 January 2021)

If we write $\Delta z = z - z_0$, then the equation $f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ can be expressed as $f'(z_0) =$ _____.

Select the correct option

○	$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$
○	$\lim_{\Delta z \to 0} \frac{f(\Delta z) - f(z_0)}{\Delta z}$
○	$\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$ ✓
○	$\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z)}{\Delta z}$

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MTH632:Grand Quiz

Quiz Start T

Question # 26 of 30 (Start time: 07:19:42 PM, 02 January 2021)

Let z = 7 i, then r = _____.

Select the correct option

- 1
- 2
- 4
- 7 ✓

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MTH632:Grand Quiz

Question # 28 of 30 (Start time: 07:20:44 PM, 02 January 2021)

If $-8i = 8 \exp[\, i\,(-\pi/2+2k\pi)\,]$, then the root of

$$(-8i)^{1/3}$$

for k=1 is _____.

Select the correct option

○	3i
○	-3i
○	2i ✓
○	-2i

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Question # 29 of 30 (Start time: 08:30:53 PM, 02 January 2021)

If $f(z) = \dfrac{4z^4 - 1}{(z^2 - 1)(z - 2)}$, then f(z) is not analytic at z =

Select the correct option

O	1 ✓
O	3
O	4
O	5

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MTH632:Grand Quiz

Quiz Start Time: 09:33 PM, 02 January

Question # 24 of 30 (Start time: 09:59:18 PM, 02 January 2021)

Total Ma

A point at which a function ceases to be analytic is called a _____ point.

Select the correct option

- regular
- non-regular
- singular ✓
- non-singular

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MTH632:Grand Quiz

Question # 24 of 30 (Start time: 11:43:34 PM, 02 January 2021)

Harmonic conjugate of $u(x, y) = e^y \cos x$ is _____.

Select the correct option

○	$e^y \sin x + c$
○	$-e^y \sin x + c$ ✓
○	$e^x \sin y + c$
○	$e^x \cos y + c$

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MTH632:Grand Quiz

Question # 25 of 30 (Start time: 11:44:05 PM, 02 January 2021)

If $f(z) = z^2$, $g(z) = \frac{1}{z}$, then $g(f(3)) =$ _

Select the correct option

O	1/9 ✓
O	9
O	3
O	1/3

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Question # 9 of 30 (Start time: 09:45:46 PM, 02 January 2021)

In z = a + bi, if i is replaced by −i, then another complex number obtained is said to be

Select the correct option

- o additive inverse of z
- o prime factor of z
- o Complex conjugate of z ✓
- o multiplicative inverse of z

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Question # 5 of 30 (Start time: 09:41:58 PM, 02 January 2021)

If $f(z) = \dfrac{z^4 + 1}{(z^2 + 1)(z - 3)}$, then f(z) is not analytic at z =

Select the correct option

O	1
O	-1
O	3 ✓
O	4

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MTH632:Grand Quiz

Quiz Start Ti

Question # 4 of 30 (Start time: 09:24:32 PM, 02 January 2021)

What is the modulus of the complex number 8i?

Select the correct option

- ● 8 ✓
- ○ -8
- ○ i
- ○ -i

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MTH632:Grand Quiz

Question # 30 of 30 (Start time: 08:31:45 PM, 02 January 2021)

The square roots of 2i are

Select the correct option

○ ✓	$\sqrt{2}\left[i\left(\frac{\pi}{4}+\frac{2k\pi}{2}\right)\right], k \in Z$
○	$\sqrt{3}\left[i\left(\frac{\pi}{4}+\frac{2k\pi}{2}\right)\right], k \in Z$

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Question # 21 of 30 (Start time: 08:26:15 PM, 02 January 2021)
Total Marks

In an Argand diagram, the complex number z = 2i will be a point _____.

Select the correct option

- 2 units along the positive imaginary axis ✓
- 2 units along the positive real axis

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MTH632:Grand Quiz

Question # 5 of 30 (Start time: 08:19:15 PM, 02 January 2021)

The principal argument of -5-5i is

Select the correct option

○	$\frac{5\pi}{4}$
○	$\frac{-3\pi}{4}$
○	$\frac{3\pi}{4}$
○	$-\frac{5\pi}{4}$

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MTH632:Grand Quiz

Question # 29 of 30 (Start time: 07:21:15 PM, 02 January 2021)

$$|z_1 - z_2| =$$

Select the correct option

○	$> |z_1| + |z_2|$
○ ✓	$\leq |z_1| + |z_2|$
○	$\leq z_1 + z_2$
○	$> z_1 + z_2$

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MTH632:Grand Quiz

Quiz Start Time: 07:05 PM, 02

Question # 25 of 30 (Start time: 07:18:50 PM, 02 January 2021)

A curve is called _____ if it does not cross itself.

Select the correct option

- ○ closed curve
- ○ open curve
- ○ simple curve ✓
- ○ parallel curve

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Question # 24 of 30 (Start time: 07:17:53 PM, 02 January 2021)

The Cartesian form of the complex number 6cis(π) is

Select the correct option

O	-6 ✓
O	6
O	6π
O	-6π

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Question # 23 of 30 (Start time: 07:17:15 PM, 02 January 2021)

If $f(z) = \frac{z^4 + 1}{(z^2 + 1)(z - 5)}$, then f(z) is not analytic at z =

Select the correct option

○	1
○	-1
○	3
○	5 ✓

$$f(z) = \frac{z^4 + 1}{(z^2 + 1)(z - 5)}$$

$$(z^2 + 1)(z - 5) = 0$$

$$z = \pm i, z = 5$$

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Question # 21 of 30 (Start time: 07:16:03 PM, 02 January 2021)

The three cube roots of the number -8i are

Select the correct option

○	$2\exp\left[i\left(-\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right],\ k=0,1,2$ ✓
○	$2\exp\left[i\left(-\frac{\pi}{6}+\frac{2k\pi}{3}\right)\right],\ k=0,1$

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MTH632:Grand Quiz

Question # 20 of 30 (Start time: 07:15:29 PM, 02 January 2021)

The polar form of the complex function $f(z) = z^2$ *is*

Select the correct option

○	$r^2(cos\theta + isin\theta)$
○	$r^2(cos2\theta + isin2\theta)$ ✓

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Question # 19 of 30 (Start time: 07:14:57 PM, 02 January 2021)

$$If\ f(z) = z^2,\ g(z) = \frac{1}{z},\ \text{then } g(f(4)) =$$

_____.

Select the correct option

○	4
○	16
○	1/4
○	1/16 ✓

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MTH632:Grand Quiz

Question # 18 of 30 (Start time: 07:14:25 PM, 02 January 2021)

The complex number

$$(4i)^4$$

have _____ number of roots.

Select the correct option

○	1
○	2
○	3
○	4 ✓

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Question # 17 of 30 (Start time: 07:13:54 PM, 02 January 2021)

The polar form of the complex function $f(z) = z^3$ *is*

Select the correct option

○	$r^2(cos3\theta + isin3\theta)$
○	$r^3(cos3\theta + isin3\theta)$ ✓

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Question # 15 of 30 (Start time: 07:12:50 PM, 02 January 2021)

The polar form of the complex function $f(z) = \frac{1}{z}$ *is*

Select the correct option

○	$r(\cos\theta + i\sin\theta)$
○	$r(\cos\theta - i\sin\theta)$
○	$\frac{1}{r}(\cos\theta - i\sin\theta)$ ✓
○	$\frac{1}{r}(\cos\theta + i\sin\theta)$

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MTH632:Grand Quiz

Question # 14 of 30 (Start time: 07:12:02 PM, 02 January 2021)

$$\textit{The value of } w = \frac{1}{z} \textit{ when } z = 2 - i \textit{ is}$$

Select the correct option

○	$w = \frac{1}{z} = \frac{1}{2-i} \times \frac{2+i}{2+i} = \frac{2+i}{2^2 - i^2} = \frac{1}{5}\ 2+i$ $\frac{1}{5}(2+i)$ ✓
○	$\frac{1}{5}(2-i)$

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MTH632:Grand Quiz

Question # 11 of 30 (Start time: 07:10:07 PM, 02 January 2021)

$$(1+i)^2 = _____.$$

Select the correct option

○	1+2i
○	0+2i ✓
○	1+i
○	1-2i

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MTH632:Grand Quiz

Question # 12 of 30 (Start time: 07:10:37 PM, 02 January 2021)

$$\lim_{(x,y)\to(1,2)} 3xy^2 - y = _____$$

Select the correct option

○	8
○	9
○	10 ✓
○	11

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MTH632:Grand Quiz

Quiz Start Time: 07:05 PM, 02 Ja

Question # 8 of 30 (Start time: 07:08:50 PM, 02 January 2021)

To

If f approaches two complex numbers $L_1 \neq L_2$ for two different curves or paths through z_0 then the $\lim_{z \to z_0} f(z)$ _____

Select the correct option

Reload Mat

- exists
- does not exist ✓

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Question # 6 of 30 (Start time: 07:07:55 PM, 02 January 2021)

The distance between two complex numbers z_1 and z_2 is

Select the correct option

○	$\lvert z_1 - z_2 \rvert$ ✓
○	$\lvert z_1 + z_2 \rvert$

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MTH632:Grand Quiz

Question # 5 of 30 (Start time: 07:07:21 PM, 02 January 2021)

Polar form of the complex number 2-2i is

Select the correct option

○	$2cis\left(\frac{\pi}{4}\right)$
○	$2cis\left(-\frac{\pi}{4}\right)$
○ ✓	$2\sqrt{2}cis\left(-\frac{\pi}{4}\right)$
○	none of these

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MTH632:Grand Quiz

Question # 27 of 30 (Start time: 06:19:07 PM, 02 January 2021)

The complex number -1-i has exponential form

Select the correct option

○	$\sqrt{2}\left[i\left(-\frac{3\pi}{2}\right)\right]$	
●	✓	$\sqrt{2}\left[i\left(-\frac{3\pi}{4}\right)\right]$

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Question # 17 of 30 (Start time: 06:12:29 PM, 02 January 2021)

A closed disc with center z_0 and radius ϵ is defined by

Select the correct option

- $|z_1 - z_0| \geq \epsilon$
- $|z_1 - z_0| \leq \epsilon$

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Question # 6 of 30 (Start time: 11:32:15 PM, 02 January 2021)

If we write the function $w = z^2 - z$ as $w = u(x,y) + iv(x,y)$, then $u(x,y) =$

Select the correct option

O	$x^2 - x - y^2$ ✓
O	$x^2 - y^2$
O	$x^2 + y^2$
O	none of these

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MTH632:Grand Quiz

Question # 2 of 30 (Start time: 11:29:36 PM, 02 January 2021)

The conjugate of 6 + 3i is

Select the correct option

- -6-3i
- -6+3i
- 6-3i ✓
- 6+3i

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Question # 1 of 30 (Start time: 11:28:30 PM, 02 January 2021)

If we write the function $w = z^2 - z$ *as* $w = u(x,y) + iv(x,y)$, *then* $v(x,y) =$

Select the correct option

O	$x^2 - x - y^2$
O	$2xy - y$ ✓

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Question # 29 of 30 (Start time: 10:04:19 PM, 02 January 2021)

If two complex - valued functions f and g satisfy conditions $g(f(z)) = z$ and $f(g(w)) = w$, then these functions are_____of each other.

Select the correct option

O	conjugates
O	inverses ✓
O	reciprocals
O	none of the above

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The fifth roots of -1 are exactly



Select the correct option

o	cis(π/5), cis(3π/5), cis((-3π)/5) and cis((-π)/5)
o	cis(± π/5), cis(±3π/5), and -1
o	cis(± π/10), cis(±3π/10), and -1
o	none

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Question # 14 of 30 (Start time: 09:50:11 PM, 02 January 2021)

For which values of x is $w = \frac{1}{x}$ defined?

Select the correct option

- w is defined for all $x \neq 0$. ✓
- w is defined for all $x = 0$.

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MTH632:Grand Quiz

Question # 9 of 30 (Start time: 11:34:13 PM, 02 January 2021)

If $z = -1+i$, then z expressed in polar form is

Select the correct option

○	$2cis\left(-\frac{\pi}{4}\right)$
○	$2cis\left(\frac{3\pi}{4}\right)$
○	$cis\left(-\frac{\pi}{4}\right)$
○	$cis\left(\frac{3\pi}{4}\right)$

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MTH632:Grand Quiz

Question # 10 of 30 (Start time: 11:34:50 PM, 02 January 2021)

The limit $\lim_{z \to 0} \frac{z}{\bar{z}}$ _____

Select the correct option

○	exists
○	does not exists ✓

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MTH632:Grand Quiz

Quiz Start Time: 11:28

Question # 12 of 30 (Start time: 11:36:09 PM, 02 January 2021)

The complex number z = 2+i0 will be located _____

Select the correct option

- at 2 on the real axis. ✓
- at -2 on the imaginary axis
- at -2 on the real axis
- at 2 on the imaginary axis

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Question # 16 of 30 (Start time: 11:38:35 PM, 02 January 2021)

The value of $w = z^2 - z$ *when* $z = 2 - i$ *is*

Select the correct option

○	$w = \frac{2}{5} + \frac{1}{5}i$ ✓
○	$w = \frac{2}{5} - \frac{1}{5}i$

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MTH632:Grand Quiz

Quiz Start Time: 11:28 PM

Question # 26 of 30 (Start time: 11:44:58 PM, 02 January 2021)

What is the modulus of the complex number -6 +8i?

Select the correct option

- 10 ✓
- -10
- 6
- 14

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Question # 22 of 30 (Start time: 05:05:26 PM, 02 January 2021)

What is the modulus of the complex number 6+0i?

Select the correct option

- O 6
- O -6
- O 0
- O None of these

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MTH632:Grand Quiz

Question # 22 of 30 (Start time: 09:33:39 PM, 02 January 2021)

Which of the following does NOT describe the modulus of the complex number a + bi?

Select the correct option

O	The length of the directed line segment from the origin of the complex plane to the point (a, b)
O	The magnitude of the vector representing the complex number a + bi.
O	$\sqrt{a^2 + b^2}$
O	The difference between the numbers a and b. ✓

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MTH632:Grand Quiz

Question # 12 of 30 (Start time: 09:33:52 PM, 02 January 2021)

In general, 1^(1/n) is

Select the correct option

●	$\exp\left(i\frac{2\pi}{n}\right)$
○ ✓	$\exp\left(i\frac{2k\pi}{n}\right),\ k=0,1,\ldots,n-1$

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Question # 17 of 30 (Start time: 09:52:51 PM, 02 January 2021)

The polar form of a complex number z is

Select the correct option

O	$r(\tan\theta + i\cot\theta)$
O	$r(\sec\theta + i\,\mathrm{cosec}\,\theta)$
O	$r(\sin\theta + i\cos\theta)$
O	$r(\cos\theta + i\sin\theta)$ ✓

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MTH632:Grand Quiz

Question # 14 of 30 (Start time: 09:27:55 PM, 02 January 2021)

If $z = a + ib$ is a complex number, then

Select the correct option

O	$\mathrm{Re}(z) = \frac{z + \bar{z}}{2i}$
O	$\mathrm{Re}(z) = \frac{z + \bar{z}}{2}$
O	$\mathrm{Re}(z) = z + \bar{z}$
O	None of these

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